Class: SE **Subject: COA LAB** **Sem-III**

Experiment No 7

Course Outcome: CO4 **Blooms Level:L4**

Aim: Analyze the functions of INT 10H video interrupts for cursor manipulation, such as hiding and reshaping the cursor.

Abstract: Design and develop an 8086 assembly language program that provides a menu-driven interface to control and manipulate the text cursor on the screen using BIOS interrupt services. The program should support operations such as changing the cursor shape (block or underline), moving the cursor in four directions (up, down, left, right), and disabling the cursor display. The user should be able to select an option by pressing a corresponding key (1–7), and any other key should terminate the program. Cursor control should be implemented using BIOS interrupt INT 10H with appropriate function calls for reading and setting cursor position and type.

Theory:

Interrupts in 8086

There are some extremely useful subroutines within BIOS or DOS that are available to the user through the INT (Interrupt) instruction.
• Format: INT xx; the interrupt number xx can be 00-FFH
– This gives a total of 256 interrupts
– Common Interrupts
• INT 10h Video Services
• INT 16h Keyboard Services
• INT 17h Printer Services
• INT 21h MS-DOS services
– Before the services, certain registers must have specific values in them,depending on the function being requested.

DOS Interrupts

- The interrupt types 20h-3Fh are serviced by DOS routines that provide high-level service to hardware as well as system resources such as files and directories
- The most useful is INT 21h, which provides many functions for doing keyboard, video, and file operations

BIOS interrupt

- BIOS interrupt calls are a facility that operating systems and application programs use to invoke the facilities of the Basic Input/Output System on IBM PC compatible computers.

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- INT 10H subroutines are burned into the ROM BIOS and are used to communicate with the computer's screen video.Manipulation of screen text/graphics can be done via INT 10H.
- Among the functions associated with INT 10H are changing character or background color, clearing the screen, and changing the location of the cursor,each chosen by putting a specific value in register AH.

INT 10H function AH = 02 will change the position of the cursor to any location.

– Desired position is identified by row/column values in DX.

• Where DH = row and DL = column.

I**NT10H function AH=03** will get current cursor position

Registers DH and DL will have the current row and column positions and CX provides info about the shape of the cursor.

INT 10h function AH = 01h will set text-mode cursor shape.
input:
CH = cursor start line (bits 0-4) and options (bits 5-7).
CL = bottom cursor line (bits 0-4).
when bit 5 of CH is set to 0, the cursor is visible. when bit 5 is 1, the cursor is not visible.
CX=2607h is an invisible cursor.

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Program:

```
.model small
.stack 100h
.data
menu db 13,10, 'case1. increase cursor (block)',13,10
     db 'case2. decrease cursor (underline)',13,10
     db 'case3. move right',13,10
     db 'case4. move left',13,10
     db 'case5. move up',13,10
     db 'case6. move down',13,10
     db 'case7. disable cursor',13,10
     db 'Press key 1-7, any other key to exit...',13,10,'$'
.code
main:
    mov ax, @data
    mov ds, ax

    ; Display menu
    mov ah, 09h
    lea dx, menu
    int 21h

main_loop:
    ; Wait for key press
    mov ah, 01h
    int 21h
    mov bl, al     ; Store key in BL

    ; Check cases
    cmp bl, '1'
    je increase_cursor
    cmp bl, '2'
    je decrease_cursor
    cmp bl, '3'
    je move_right
    cmp bl, '4'
    je move_left
    cmp bl, '5'
    je move_up
    cmp bl, '6'
    je move_down
    cmp bl, '7'
    je hide_cursor

    jmp exit_program
```

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```
increase_cursor:
    mov ah, 01h
    mov ch, 00h     ; Start line (top)
    mov cl, 07h     ; End line (bottom) - block
    int 10h
    jmp main_loop

decrease_cursor:
    mov ah, 01h
    mov ch, 06h     ; Underline
    mov cl, 07h
    int 10h
    jmp main_loop

move_right:
    mov ah, 03h
    mov bh, 0
    int 10h         ; Get current position
    inc dl          ; Move right
    mov ah, 02h
    int 10h
    jmp main_loop

move_left:
    mov ah, 03h
    mov bh, 0
    int 10h
    dec dl
    mov ah, 02h
    int 10h
    jmp main_loop

move_up:
    mov ah, 03h
    mov bh, 0
    int 10h
    dec dh
    mov ah, 02h
    int 10h
    jmp main_loop

move_down:
    mov ah, 03h
    mov bh, 0
    int 10h
    inc dh
    mov ah, 02h
    int 10h
    jmp main_loop

hide_cursor:
    mov ah, 01h
    mov ch, 26h    ; Invalid shape (hide)
    mov cl, 07h
    int 10h
    jmp main_loop

exit_program:
    mov ah, 4Ch
    int 21h
end main
```

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Output:

```
C:\>EXP7.EXE

case1. increase cursor (block)
case2. decrease cursor (underline)
case3. move right
case4. move left
case5. move up
case6. move down
case7. disable cursor
Press key 1-7, any other key to exit...
1
```

```
C:\>EXP7.EXE

case1. increase cursor (block)
case2. decrease cursor (underline)
case3. move right
case4. move left
case5. move up
case6. move down
case7. disable cursor
Press key 1-7, any other key to exit...
12
```

```
C:\>EXP7.EXE

case1. increase cursor (block)
case2. decrease cursor (underline)
case3. move right
case4. move left
case5. move up
case6. move down
case7. disable cursor
Press key 1-7, any other key to exit...
3
```

```
C:\>EXP7.EXE

case1. increase cursor (block)
case2. decrease cursor (underline)
case3. move right
case4. move left
case5. move up
case6. move down
case7. disable cursor
Press key 1-7, any other key to exit...
3 4
```

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```
C:\>EXP7.EXE

case1. increase cursor (block)
case2. decrease cursor (underline)
case3. move right
case4. move left
case5. move up
case6. move down
case7. disable cursor
Press key 1-7, any other key to exit...
3 5
```

```
C:\>EXP7.EXE

case1. increase cursor (block)
case2. decrease cursor (underline)
case3. move right
case4. move left
case5. move up
case6. move down
case7. disable cursor
Pre6s key 1-7, any other key to exit...
3 5 6

       _
```

```
C:\>EXP7.EXE

case1. increase cursor (block)
case2. decrease cursor (underline)
case3. move right
case4. move left
case5. move up
case6. move down
case7. disable cursor
Pre6s key 1-7, any other key to exit...
3 5 6

       7
```

Conclusion:

This program successfully implements a menu-driven interface for controlling the text-mode cursor using 8086 assembly language. By leveraging BIOS interrupt INT 10H functions, it enables the user to dynamically adjust the cursor shape, position, and visibility. The use of keyboard input and direct cursor manipulation demonstrates a practical application of low-level programming concepts, such as register usage, interrupt handling, and hardware-level screen control.

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Exercise 1:

Aim: To analyze and implement BIOS interrupt services in 8086 assembly language for drawing basic geometric shapes such as rectangles and triangles

Abstract: Design and implement an 8086 assembly language program that utilizes BIOS video interrupt services to draw basic geometric shapes on the screen in graphics mode. The program should set the display to video mode 13h (320x200 pixels with 256 colors), draw a rectangle and a triangle using pixel plotting techniques, and wait for a key press before restoring the screen to text mode. The rectangle should be drawn using horizontal and vertical lines, while the triangle should include a horizontal base and two slanting lines forming the sides. The program should include a subroutine to plot individual pixels at specified coordinates.